

Pandas in Python

1. What is Pandas?

Pandas is a Python library used for data analysis and data manipulation.

It helps us work with data in tables, such as CSV files, Excel files and databases.

2. Why Use Pandas?

- Read and write data files.
- Organize data in tables.
- Clean missing or incorrect data.
- Filter and sort information.
- Analyze and summarize data.

3. Main Data Structures.

Data Structure

Meaning

Series

One-dimensional data

(single column)

Data frame

Two-dimensional data

(rows and columns)

Dataframe is most commonly used in Pandas.

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4. Important Pandas Concepts

Series - A single column of data with label (index)

Dataframe - A table containing rows and columns

Index - Labels used to identify rows

Columns - The named fields in a Dataframe.

Missing values - Data that is empty or unavailable

5. Common Pandas Functions.

Function	Purposed
<code>read_csv()</code>	Read a csv file
<code>head()</code>	Show the first rows
<code>tail()</code>	show the last rows
<code>info()</code>	Display data information
<code>describe()</code>	Provides statistical summary
<code>shape</code>	Shows number of rows and column
<code>columns</code>	shows column names
<code>fillna</code>	Fill missing values
<code>drop()</code>	Removes rows or columns
<code>isnull()</code>	Check missing values.

6. Reading and Writing File.

- CSV
- Excel
- JSON
- SQL Databases.

7. Selecting Data.

Pandas provides ways to select data from a DataFrame.

Column selection: Selects one or more columns

Row selection: Selects specific rows.

loc: Selects data using labels

iloc: Selects data using integer positions

8. Data Cleaning.

Data cleaning means fixing or removing incorrect missing or duplicate data.

Common operations.

- Handling missing values
- Removing duplicate rows
- Changing data types
- Removing un.
- Correcting column names
- Removing unnecessary spaces
- Replacing incorrect values.

9. Filtering Data

Filtering means selecting rows that meet a specific condition.

It helps us find particular information from a large dataset.

For example, finding products with a price greater than a certain amount.

10. Sorting data

Sorting means arranging data in a specific order.

Types.

- Ascending order (small to large)
- Descending order (large to small)

11. GroupBy

GroupBy is used to divide data into groups based on a column and perform calculations on each group.

Commons Operations.

- Count
- Sum
- Average
- Minimum
- Maximum

12. Merging and Joining

Pandas can combine data from Multiple Dataframes.

- Merge :- Combines tables using common columns or Keys.
- Concat :- Combines DataFrames along rows and columns.
- Join :- Combines data based on indexes or Keys

13. Important Pandas Methods.

Method	Purpose
sort_value()	Sorts data
value_counts()	Counts unique values
groupby()	Groups data
drop_duplicates()	Remove duplicates
rename()	Changes column names
dropna()	Removes missing data
fillna()	Fill missing data
apply()	Applies a function to data

14. Applications of Pandas

- Data analysis
- Business reporting
- Financial data analysis
- Data cleaning
- Machine learning preparation
- Working with CSV datasets
- Data visualization preparation.

Quick Revision

Pandas = Data Analysis + Data Cleaning
+ Data Manipulation.

In short :- Pandas helps us read, clean, organize and analyse data easily in Python.